Data plan: one consistent dataset, read one way (v2)

Owner, 2026-10-23: “Where is the plan for more dilligent data collection and consisten analysis? … the mismatch in what data we are using for various parts of the model is entirely haphazard and inconsistent.”

Evidence: `INVENTORY.md` (what is held) and `DATA-USE-AUDIT.md` (what each part of the model reads; its 17 inconsistencies are A1–A17, its open questions Q-a…Q-k). v1 was reviewed in `DATA-PLAN-REVIEW.md`; v2 applies all 24 of its changes. `PLAN.md` (the backtest round) carries out its steps 1–3 through this plan. Nothing here is built until the owner has read it.

1. The rule

Every part of the model reads the same facts, defined once, from one store, through one gate that knows the date the forecast is made as of. A second reader of a fact, or a second definition of it, is a defect, and a test finds it.

2. Definitions

Term	Definition
Ballot	LGE ward, LGE PR, NPE national, NPE provincial, and for 2024 the compensatory national and the regional ballot, each named. Nothing is called “national” that is the provincial ballot (A6).
Votes cast	the source’s `TOTAL VOTES CAST` where it has one, else valid + spoilt; `special` and `s24a` carried beside it where the source has them.
Turnout	votes cast ÷ registered, per ballot. One turnout of record per election kind: the PR ballot for an LGE, the national ballot for an NPE. At VD level, net of s24a votes where held (s24a voters cast outside their VD).
Share, contested share, combined vote	share = party valid votes ÷ valid votes on one ballot; contested share = share among the wards a party stood in; combined vote = ward + PR valid votes, the statute’s basis for seats.
Roll	each election’s quantities on that election’s own roll. One target roll per run, dated and recorded. A past target’s roll and VD→ward map enter the store from a reviewed extract dated by the roll’s certification (C15); the vote rows are not opened to build them.
Geography	quantities carried at VD level, placed in wards by one join, in the delimitation of the election they describe; moved to another delimitation by one crosswalk with weights and a linked-share figure; one rule for ward→VD downscaling and one for VDs split across wards. Backtests cannot exercise split VDs (no past file has one); stated wherever a backtest is quoted.
Party identity	one table of codes, names by date and events (rename, merger, split), each with a source public before the target that uses it; independents identified per candidate.
Census	the latest census released before the target; Census 2022 (the owner’s exception) only where none is held; labelled at every output it reaches.

Term	Definition
Period	every fact carries `period_end`, `public_on` and a vintage. Every run carries `as_of`: polling day for a backtest, the freeze date for the forecast. **Clean** = `public_on` < `as_of`. A pre-election fact held in a result file (roll, ward map, ballot presence) is stored in its own table with its own `public_on`.
Excluded data	`defective` (excluded for every reader, permanently, with the defect) is separate from `withheld for diagnosis` (an arm, applied to every reader or none) (A1).
Window	each quantity's window of years declared with its reason in `JUDGEMENT-CALLS.md`; a window chosen by score is chosen inside the rolling origin (A8).

3. The store

One generator, `src/store.py`, builds `data/store/` from `data/raw/` only. Its rebuild is a re-emit window under `CLAUDE.md` §3 (one writer; nobody measures while it runs).

Table	One row per
`results`	election × ballot × VD × party: votes
`vd_totals`	election × ballot × VD: registered, valid, spoilt, cast, special, s24a
`rolls`	roll vintage × VD or ward: registered, certified on
`vd_ward`	delimitation × VD × ward part: weight, source
`vd_links`	VD in election A × VD in election B: method, confidence
`nominations`	election × ward × party: stood
`byelections`	contest × VD (× party): votes; registered, spoilt, cast as fetched
`census`	census × delimitation × ward × group (and age): persons, released on
`parties`	party × event: code, name, event, date, source
`statute`	election × municipality: council size, wards, formula, overhang rule, source
`municipal_extent`	municipality × period: area changes, source
`polls`	poll: house, mode, frame, fieldwork dates, published on, ballot asked (or a statement that polls stay out of the model)

Every fact is keyed by vintage. Each table has a checksum and a manifest entry (generator commit, inputs, checksums).

The gate. `store.view(city, as_of)` returns only clean facts, plus the labelled census exception. Every model part gets its data from a view; the view records what it returned, and that record is part of every run.

Derived artefacts (pool specs, `turnout.csv`, fold parameters, width record, by-election weights) record the views they were built from; their `period_end` and `public_on` are the maximum over their inputs, computed by the writer; a loader refuses an artefact whose views do not match the run.

Truth for scoring is `store.truth(city, T)`, read only in a separate process after the forecast's artefacts are hashed.

What the gate cannot see, and how each is covered: constants and lists in code (`PLAN.md` 3.1's classification); a hand-typed `public_on` (each comes from a cited source, reviewed); 2026 geometry (dated as a 2026 fact); revised vintages (kept, not overwritten).

4. Collection

Johannesburg first, then the other seven. Each item ends with data in `data/raw/` with provenance in `SOURCES.md`, or the exact reason it cannot be had. **A collection ledger** (`rounds/2026-10-23-out-of-sample/COLLECTION-LEDGER.md`) records every attempt: item, date, who, source URL, result. Nothing replaces a file the model reads until an approved change does so with an old-vs-new report; fetches go to a separate directory.

#	Item	Source	Check
C1	NPE national ballot 1999–2014	zips on disk	party votes sum to valid votes
C2	NPE national ballot 2019; 2024 compensatory national	IEC downloads	as C1
C3	Votes cast, special votes, every election	zips	cast against valid + spoilt, differences listed
C4	Ward for NPE 1999 (via LGE 2000), 2004, 2019, 2024 (via the preceding LGE); one VD→ward resolver for every reader (fixes `national_fit_refused` at 2021, A4)	derived	share of registration assigned
C5	By-election registration, spoilt, cast	dashboard contest files	every contest, both cycles
C6	By-elections before 2016	dashboard serves 2016 and 2021 only (404 otherwise, 2026-10-23); IEC media releases, municipal notices	or a reason per cycle
C7	By-elections after 2026-08-07; JHB 2019-07 to 2021-11	dashboard	against C22
C8	Census 2011 and 2001 by ward, population group and age	SuperWEB2 first; owner sends a request for what it lacks	ward counts per delimitation
C9	Ward polygons 2000, 2006	MDB	ward counts
C10	2026 rolls for the six metros without one; Tshwane's into its crosswalk; a Johannesburg refresh if the certified roll is published after 2026-09-26	IEC ward page	against the IEC municipal figure
C11	Certified 2026 rosters, six metros	IEC	`complete = true` only from the certified list
C12	Party events with dated sources	IEC register, press	source before the next election
C13	NPE 2024 regional ballot	IEC	as C1

#	Item	Source	Check
C14	s24a votes per VD, every NPE where held	zips	per VD
C15	Roll certification date per election; certified municipal totals 2011, 2016, 2021	IEC releases, Government Gazette	against the result files' registration
C16	Community Survey 2016, municipality level	Stats SA	citywide check for 2016
C17	Statute per election	gazette, IEC worked examples	reproduces the IEC seat calculation
C18	Municipal extent changes 2000–2026 (Tshwane 2011, Mangaung 2016 at least)	MDB	common-area citywide series
C19	Reconcile IEC 862/796 VDs against crosswalk 865/774 (JHB/TSH)	IEC, MDB	each difference explained
C20	Voting-age population by ward, each census held	Stats SA	splits registration change
C21	Poll metadata, if polls stay in the model	publishers	fieldwork and publication dates machine-readable
C22	A stored list of held by-elections per cycle, independent of the dashboard	IEC, municipal notices	C6/C7 completeness

5. Consistent analysis

The analysis tables are the model's records. θ, ρ, turnout and arrival records are read from them, so the analysis and the model share one implementation (no fourth reader of retention).

5.1 Every pair (LGE→LGE; NPE→LGE and LGE→NPE on the national and the provincial ballot), citywide and per ward, the same code for all eight metros: votes, vote change, share, retention, turnout, registration change (split into voting-age population growth and registration rate where C20 is held); the exact split of vote change into registration, turnout and share; where the votes went, net, with Duncan–Davis bounds per cell, gross flows labelled ecological estimates; population-group estimates labelled as ecological on the composition of the census used; how the changes change across consecutive pairs; by-elections against the elections either side and the correlation with the next. Ward comparisons across delimitations report the linked share and are labelled as subject to the areal-unit problem. Citywide pairs across a municipal extent change are computed on a common area or marked. A cell that cannot be computed says why. 5.2 Which national ballot predicts the local result better, on pre-target pairs only.

6. Moving the model onto the store

One change per row, written and reviewed by different agents, pre-registered where it moves a number.

Audit	Resolution
A1	`defective` and `withheld` split; the withheld-files effect (in-sample, withdrawn) re-measured on the store with the gate on every reader

Audit	Resolution
A2	one target roll per run; one split-VD method; backtests stated not to exercise split VDs; a pre-election-roll arm where C15 holds a roll
A3	one turnout of record per election kind; pool and VD turnout derived from one table; trace first (Q-a: which turnout model decides seats)
A4	latest census released before T; one VD→ward resolver (C4)
A5	owner decision with arms (fold 1; fold 2; folds 1+2 pooled), keeping the code's validation reason in view; one γ fallback rule for every city, fitted in the direction it is applied; every fold file regenerated from one commit
A6	labels now; the choice of ballot after 5.2
A7	2016–21 file as the 2021 target's input; `bye_value.py` target-aware; fetched registration (C5); the literal `share_2021` column at `montecarlo.py:6628` made target-aware
A8	each window declared with its reason; score-chosen windows inside the rolling origin
A9	benchmarks read the model's turnout table; margins re-measured
A10	`nominations` only
A11	universe from the target delimitation; measured not to reach JHB 2026 (0 of 865 VDs, 0 of 2,417,915 voters absent from NPE 2024; review, 2026-10-23)
A12	Tshwane built from the same inputs as Johannesburg before publication
A13	hard-coded paths removed; each reader run for a second city in a test
A14	a leak, not a definition: the backtest's contested share from the view at `as_of`; `NATIONAL_VOTES` and `PROJECTED_METRO_SHARE` move into the store with dates
A15	statute per election from C17
A16	one citywide-share reader
A17	`parties` replaces both mappings
Q-a … Q-k	each of the audit's open questions gets an item or a stated reason to leave it (listed when the store work starts)

Tests (each with its population, a two-sided bound, and a planted violation that fails and clears): (1) no model module opens `data/raw/` or the store except through `store.view` / `store.truth`; (2) each defined quantity has one implementation — detection by a call-graph scan of the readers, or labelled argued if no scan can see it; (3) a view at `as_of` holds no fact with `public_on` at or after it except labelled census rows; (4) every run's record lists its views.

7. Order, and what the frozen forecast carries

The 2026 forecast frozen by 31 October carries A2, A3 and A7 as they are, and A5 as the owner chose on 2026-10-23: γ from folds 1 and 2 pooled, with the fold files' placeholder rows no longer

read (OWNER-DECISIONS-ANSWERED 33, DATA-QUALITY 21). A1 and A11 do not reach it (A11 measured). Readers see the list of inputs the forecast reads, and `inputs_preflight --strict`'s output, beside the hash.

When	Work
now (no model change)	C8's SuperWEB2 attempt; C5/C7 fetches into a separate directory; the A6 labels; the A3 trace; the national–provincial gap per party for 2009 and 2014 from the zips; the A2 roll check; the collection ledger
before 31 Oct, owner's choice	old-vs-new sensitivity arms: A7 (by-election terms off) and A5 (fold 2; folds 1+2 pooled), each reported to the owner, the choice in `OWNER-DECISIONS.md`. Default if he does not choose: freeze on current settings (`PLAN.md` 5.1)
by 31 Oct	freeze; publish the hash, the input list and the preflight output
after 4 Nov	the store; C1–C4, C6, C9–C22; section 5; section 6 row by row; `PLAN.md` step 3 on the store, Johannesburg first